

DISSSI 2026 YEARLY WORKSHOP

Contributions to Integrating Deep Learning-
Based Prediction and Market Analysis in Cyber-
Physical Systems

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3-Minute Thesis

Data Science for Energy Market Analysis and Cyber-Physical Systems

The growing relevance of climate and data in the energy transition

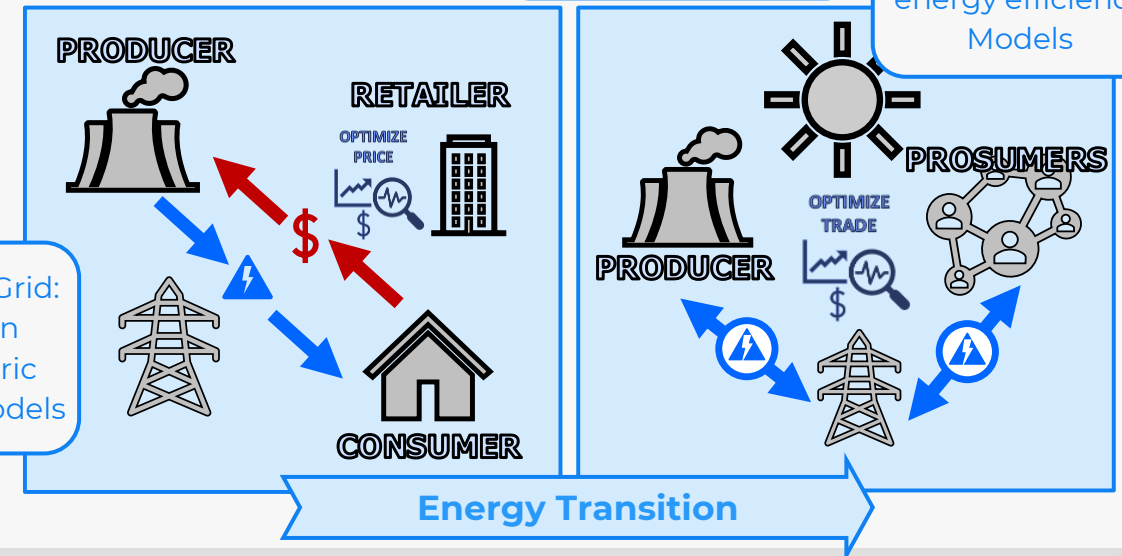


The "prosumer": challenges and opportunities in the face of an energy paradigm shift



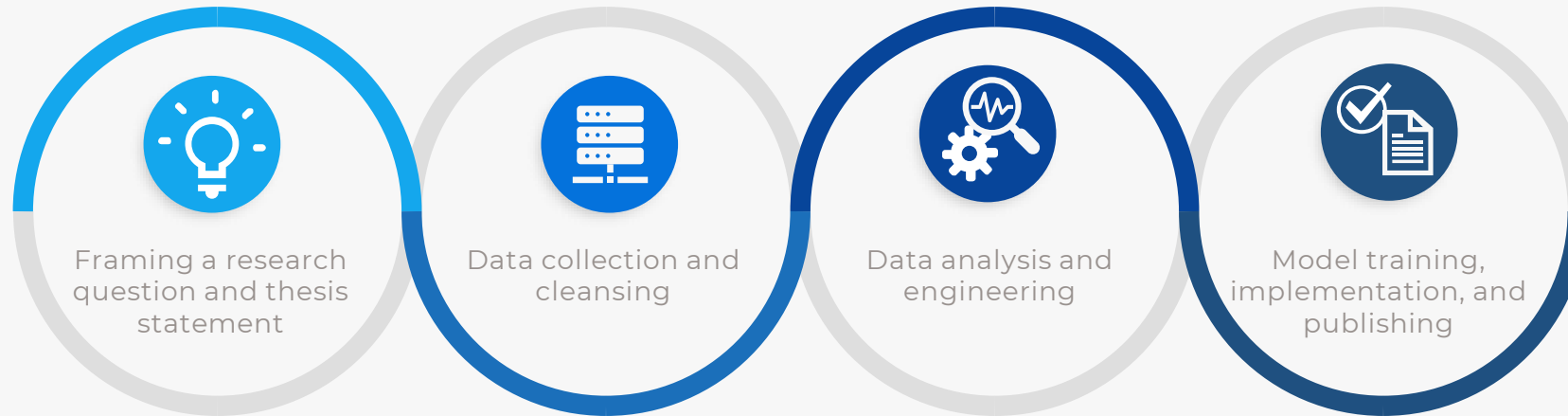
A Smart Grid of Prosumers: new energy efficiency Models

Traditional Grid: built upon econometric business models



Work Methodology

Our Ongoing Research Spans Four Core Steps of the Data Science Lifecycle



01. THESIS

The growing interdependence of weather and energy prices amidst the Prosumer and AI revolutions

02. DATA COLLECTION

- Worldwide Electricity Load Dataset (WELD)
- Global Day-Ahead Electricity Price Dataset (DAEP)
- Spain Weather Features Dataset
- World Weather Features Dataset

03. DATA ANALYSIS

- Feature engineering for historical weather variables
- Actual series vs weather anomaly series
- Time-series analysis, correlations, moving averages...

04. ML AND DL MODELS

- Electricity price forecasting
- Applications in cyber-physical systems
 - ❖ Peer-to-Peer Energy Trading
 - ❖ Smart metering algorithms...

Global Electricity Price and Load Datasets

- Published in [Mendeley Data](#) and [IEEE DataPort](#).
- Papers published in [IEEE Data Descriptions](#)
 - Describe the **data collection methodology**
 - Descriptive analysis & benchmarking of **machine learning and deep learning models**
- 10 years of data (2015–2025), 36 countries and 42 markets, 40+ million records each, 15-minute to 1-hour granularities
- Penn State Harrisburg & UPM collaboration**

Country	Data Platform	Start Timestamp	End Timestamp	Data Frequency	Data Unit	Data Collection
Japan	Japanese Power	2016-04-01T00:00:00	2023-04-02T23:00:00	1 h		[17]
Singapore	NEMS	2020-01-01T00:00:00	2025-06-02T23:30:00	30 min		[18]
Austria			2025-06-09T20:15:00	15 min		
Belgium			2025-06-09T20:30:00	15 min		
Bulgaria			2025-06-09T16:00:00	1 h		
Croatia			2025-06-09T20:30:00	1 h, 15 min		
Czech Republic			2025-06-09T20:45:00	1 h, 15 min		
Denmark			2025-06-09T20:00:00	1 h		
Estonia			2025-06-09T20:00:00	1 h		
Finland			2025-06-09T20:45:00	1 h, 15 min		
France			2025-06-09T20:45:00	1 h, 15 min		
Germany			2025-06-09T20:45:00	15 min		
Greece			2025-06-09T21:00:00	1 h		
Hungary			2025-06-09T22:15:00	15 min		
Ireland			2025-06-09T21:30:00	30 min		
Italy			2025-06-09T21:30:00	1 h, 15 min		
Latvia			2025-06-09T21:00:00	1 h		
Lithuania			2025-06-09T22:30:00	1 h, 15 min		
Luxembourg			2025-06-09T22:15:00	15 min		
Montenegro			2025-06-09T21:00:00	1 h		
Netherlands			2025-06-09T22:15:00	15 min		
North Macedonia			2025-06-09T23:00:00	1 h		
Norway			2025-06-09T21:45:00	1 h, 15 min		
Poland			2025-08-05T00:30:00	1 h, 15 min		
Portugal			2025-06-09T22:00:00	1 h		
Romania			2025-06-09T23:00:00	1 h, 15 min		
Serbia			2025-06-09T22:00:00	1 h		
Slovakia			2025-06-09T22:00:00	1 h		
Slovenia			2025-06-09T22:00:00	1 h		
Spain			2025-06-09T23:30:00	1 h, 15 min		
Sweden			2025-06-09T21:00:00	1 h		
Switzerland			2025-06-09T22:00:00	1 h		
United Kingdom						
	CAISO					
	ERCOT					
	ISO New Eng					
United States	MISO					
	NYISO					
	PJM					
	SPP					
Canada	IESO					
Australia	AEMO					

Note: Including data availability, see

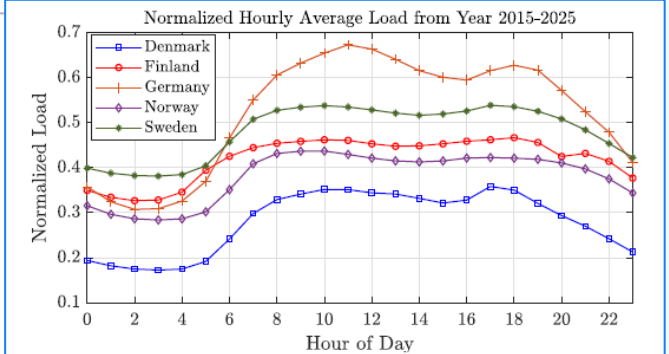
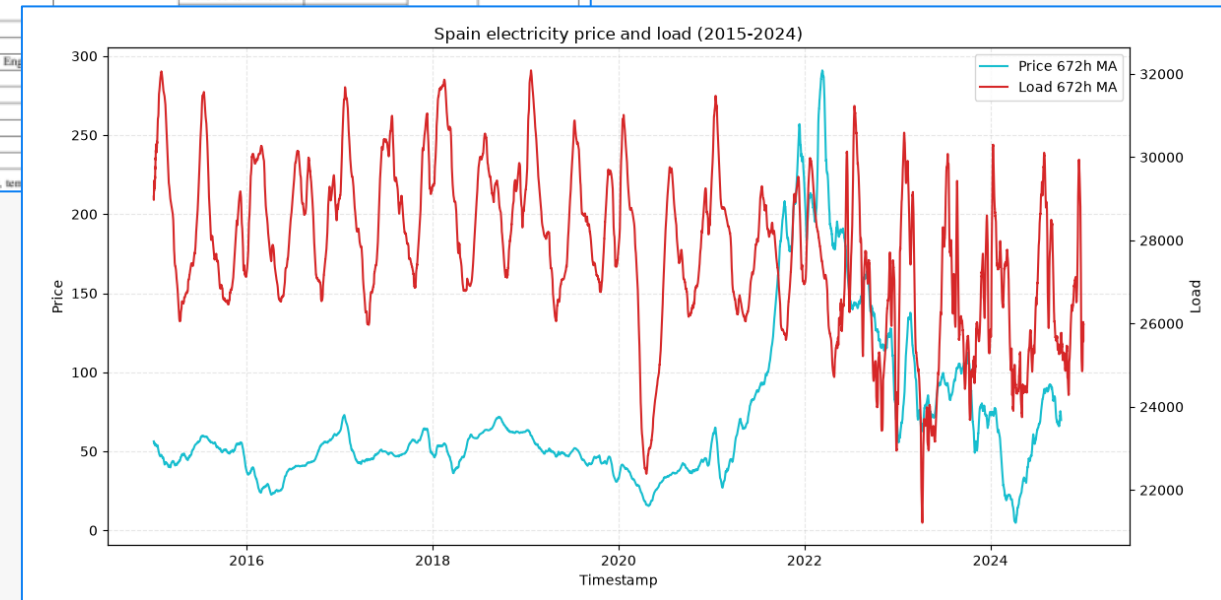
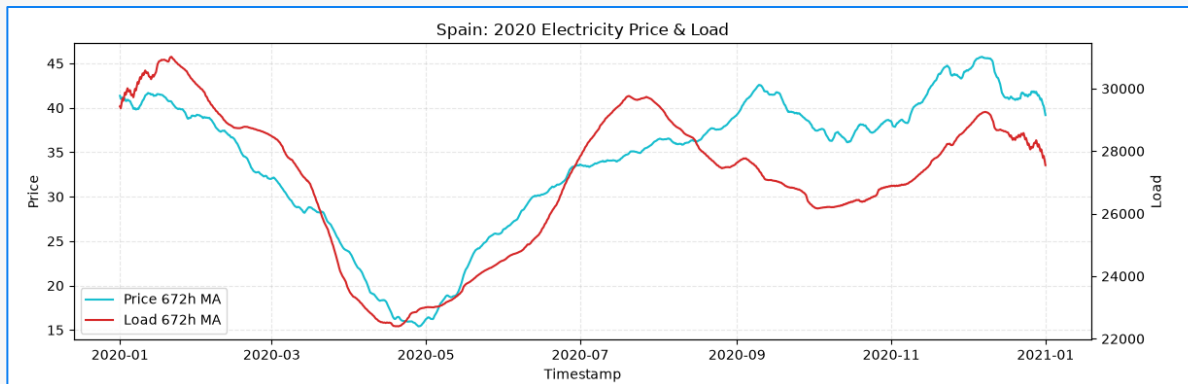


FIG. 7. Normalized hourly average electrical load in different European countries (Years: 2015-2025).

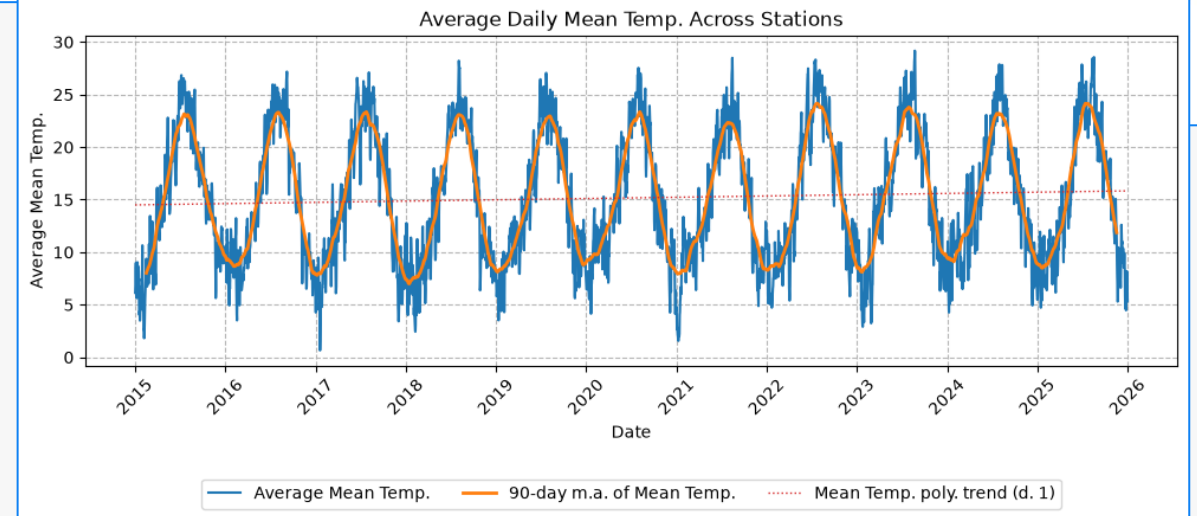
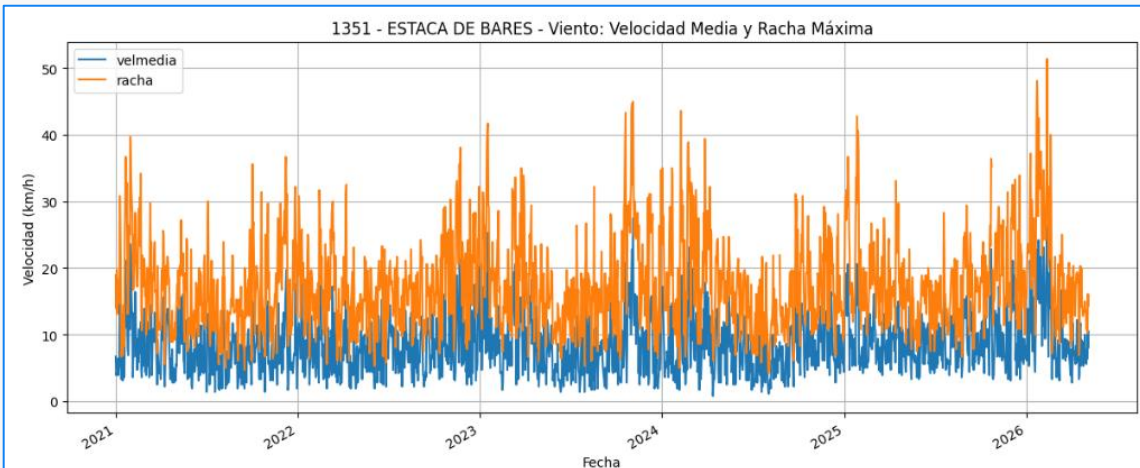
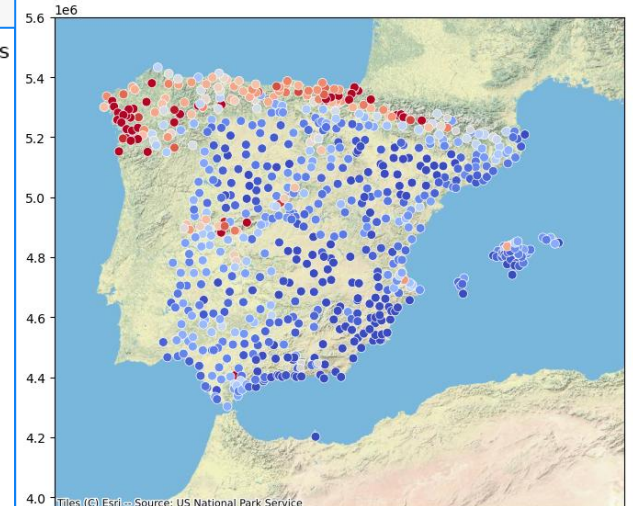
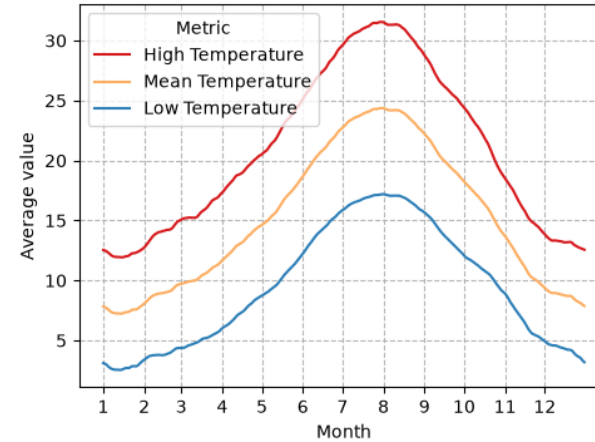


Spain Weather Features Dataset

- Dataset sourced from AEMET API, collected over the past course
- Open-sourced [code and analysis notebooks](#) 
- Distribution and Data Descriptions planned at the **IEEE Data Descriptions** journal
- 800+ weather stations across Spain, 10 years of data, 19 features, **50+ million data points**
- Multi-country version planned for next course

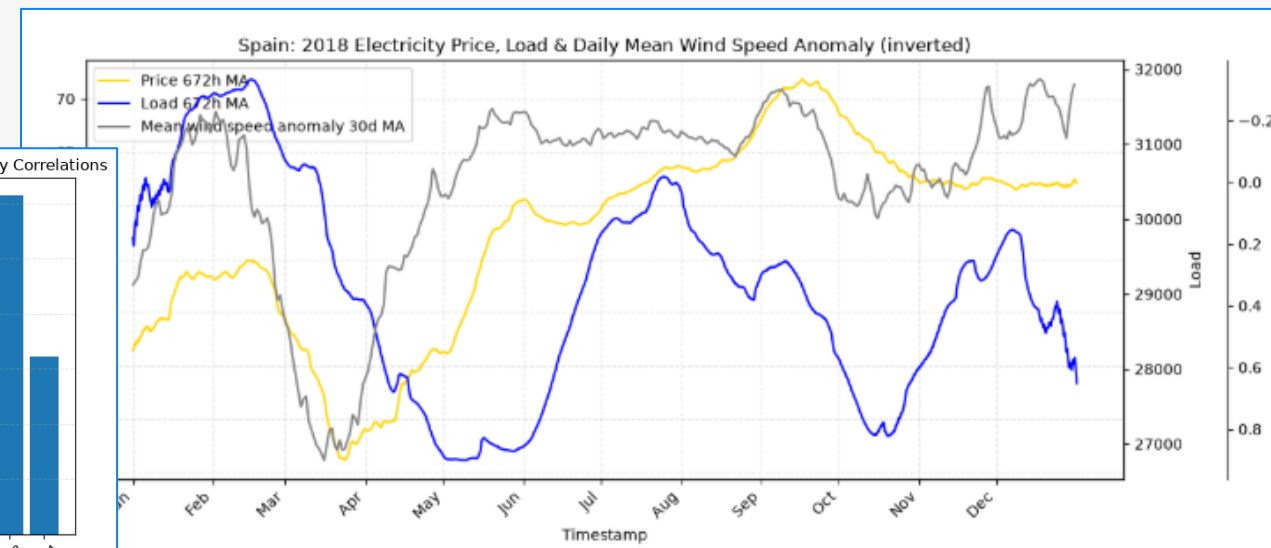
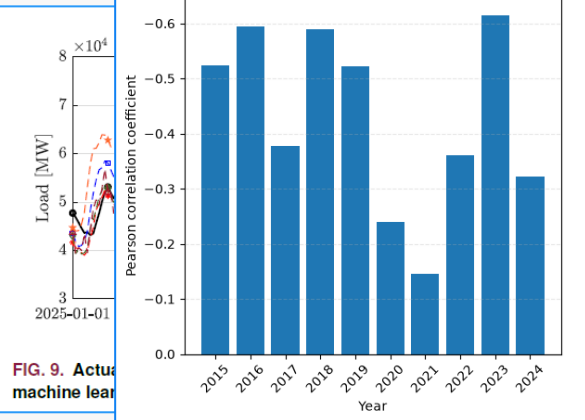
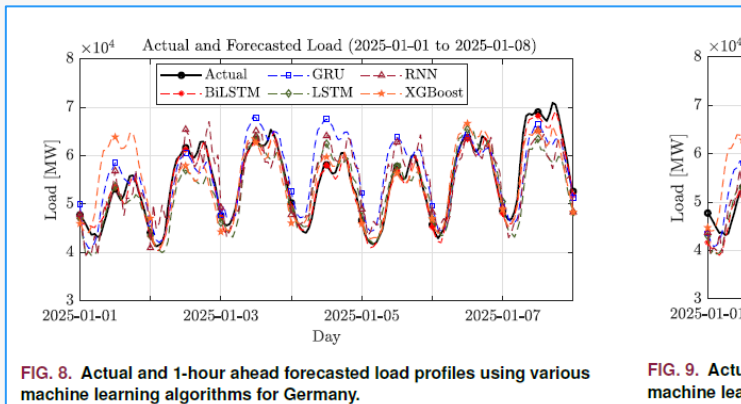
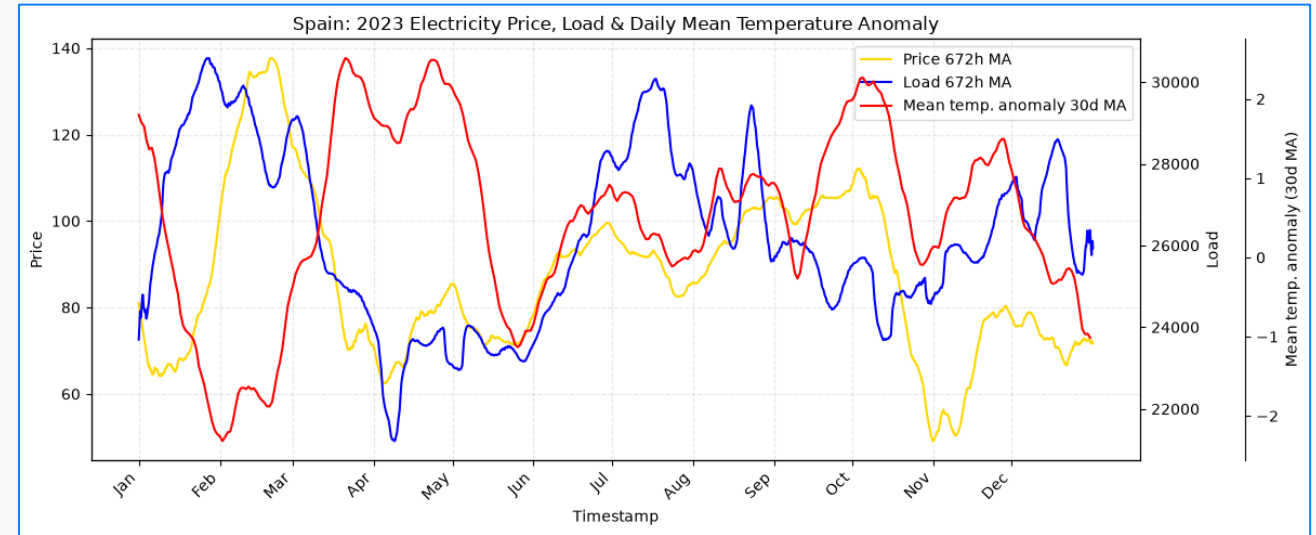
github.com/reddgr/energy-price-forecasting-in-power-grids 

30-day rolling averages of daily temperature values across stations



Transformer-BiLSTM DL Framework for Day-Ahead Energy Price Forecasting

- Published in Mendeley Data and IEEE DataPort
- Original paper by **Penn State Harrisburg** team
- Future publishable enhancements of the methodology involve **incorporating weather features** as exogenous data to complement time-series-based predictions
 - Future article ideation in progress.





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Thank you!

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